

Robert Casey Lafferty

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github.com/rcLafferty

rcLafferty.github.io

Gameplay programmer with experience in AAA and indie studios, specializing in gameplay mechanics, systems, and AI.

Game Credits

AAA: Call of Duty: Black Ops 6 (Proprietary Engine)	Oct 2024
AAA: Call of Duty: Modern Warfare III (Proprietary Engine)	Nov 2023
Indie: Social Sidekick (Unity Engine)	March 2023

Computer Skills

Languages	C++, C#, Java, C, Python, Lua, MySQL, SQL Server
Development Tools	Unreal Engine 5, Unity Engine, Git, Perforce, PlasticSVN, Audacity, Mixamo
Game AI Skills	UE State Trees, UE Behavior Trees, UE EQS, UE Perception Systems, State Machines

Professional Work Experience

Game Programming Instructor (Unreal Engine 5 C++, Unity C#, LOVE Engine Lua) Academy of Art University, Remote	Aug 2024 – Present
- Taught students crucial game development skills such as data structures, algorithms, and project management - Mentored students in software engineering practices applied to gameplay systems development	
Gameplay Engineer – AI (Unreal Engine 5 C++/Blueprint) Deviant Games (Contract), Remote	March 2024 – Jan 2025
- Architected gameplay AI subsystems coordinating NPC behaviors across multiple gameplay scenarios - Introduced lightweight peer code review practices to improve code quality and cross-team knowledge sharing	
Gameplay Engineer – Call of Duty (C++, Python, Proprietary Tech) Treyarch, Playa Vista, CA	Oct 2022 – Sept 2023
- Engineered 22 multiplayer gameplay mechanics to empower player experiences in the Call of Duty engine - Optimized workflow for my team and collaborators by centralizing information about a proprietary tool	
Software Engineer II – Tools & UI (C#, C++, MS SQL Server) AGS, Duluth, GA	Dec 2021 – July 2022
- Designed a brand new Windows service to assist in legacy support for older servers - Partnered with QA to improve UI and backend tooling, streamlining testing workflows for existing software suites	
Software Engineer – UI (C++) Naval Surface Warfare Center, Dahlgren, VA	March 2021 – Sept 2021
- Implemented UI and backend enhancements to mission-critical C++ code for government weapons systems - Streamlined new hire onboarding documentation and exercises to improve future new hire experiences	

Game Dev Projects

Resourceful (Unreal Engine 5 C++/Blueprint) – Side Project	Oct 2023 – May 2025
3D top-down fantasy game where the player manages resources to build a defense against hordes of enemies - Architected resource ecosystem and AI worker roles supporting diverse combat and progression strategies	
Wrath of Raoden (Unity C#) – Side Project (Link)	Dec 2020 – Feb 2024
2D roguelike dungeon crawler where the player must conquer randomized dungeons to save the world - Engineered mobile game from scratch with a focus on character combat, AI, and technical direction	

Education

University of Nebraska-Lincoln , Lincoln, NE Master of Science in Computer Science CS GPA: 3.39	Dec 2020
Kansas State University , Manhattan, KS Bachelor of Science in Computer Science Minor: Music CS GPA: 3.26	May 2017

Honors & Activities

Co-Founder, President – UNL Game Development Club	Aug 2018 – Dec 2019
Eagle Scout	Sept 2012